

Lecture 22

Nonparametric regression

Arnab Hazra

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Flexible regression modeling

- ▶ Nonparametric (NP) methods attempt to analyze the data by making the fewest number of assumptions as possible.
- ▶ NP methods are generally more robust and flexible, but less powerful than correctly specified parametric models.
- ▶ Most frequentist NP methods completely avoid specifying a model.
- ▶ For example, a rank or sign test to compare two means.

Non- and Semi-parametric modeling

- ▶ Bayesian methods need a likelihood in order to obtain a posterior, so you can't completely avoid specifying a model.
- ▶ Bayesian NP (BNP) then attempts to specify a model that is so flexible that it almost certainly captures the true model.
- ▶ One definition of the BNP model is one that has infinitely-many parameters.
- ▶ In some cases, NP models are difficult conceptually and computationally, and so semiparametric models with a large but finite number of parameters are useful approximations.

Parametric simple linear regression

Consider the classic parametric model:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \quad \text{where} \quad \varepsilon_i \sim N(0, \sigma^2).$$

Assumptions:

1. ε_i are independent
2. ε_i are Gaussian
3. The mean of Y_i is linear in X .
4. The residual distribution does not depend on X

Alternatives:

1. Parametric alternatives such as a time series model.
2. Let $\varepsilon_i \sim F$, and place a prior on the distribution F .
3. Let $E(Y|X) = g(X)$ and put a prior on the function g .
4. Heteroskedastic regression $\text{Var}(\varepsilon_i) = \exp(\alpha_0 + \alpha_1 X)$.

In 2-4 we are placing priors on functions, not parameters.

Nonparametric regression

- ▶ Let's relax the assumption of linearity in the mean.
- ▶ The mean is $g(X)$, where g is some function that relates X to $E(Y|X)$.
- ▶ Parametric models include
 1. Linear: $g(X) = \beta_0 + \beta_1 X$
 2. Quadratic: $g(X) = \beta_0 + \beta_1 X + \beta_2 X^2$
 3. Logistic: $g(X) = \beta_0 + \beta_1 \frac{\exp[\beta_2 + \beta_3 X]}{1 + \exp[\beta_2 + \beta_3 X]}$.
- ▶ NP regression puts a prior on the curve $g(X)$, rather than the parameters $\beta_1, \dots, \beta_p$ that determine the parametric model.

Semiparametric regression

- ▶ Semiparametric regression approximates the function g using a finite basis expansion

$$g(X) = \sum_{j=1}^J B_j(X) \beta_j$$

where $B_j(X)$ are known basis functions and β_j are unknown coefficients that determine the shape of g .

- ▶ Example: the cubic spline basis functions are

$$B_j(X) = (X - v_j)_+^3$$

where v_j are fixed knots that span the range of X .

- ▶ Many other expansions exist: wavelets; Fourier, etc.
- ▶ Fact: A basis expansion of J terms can match the true curve g at any J points $X_1, \dots, X_J$.
- ▶ So increasing J gives an arbitrarily flexible model.

Model fitting

- ▶ The model is $Y_i \sim N(B_i^T \beta, \sigma^2)$, where $\beta_j \sim N(0, \tau^2)$ and B_i is comprised of the known basis functions $B_j(X_i)$.
- ▶ Therefore, the model is usual linear regression model and is straightforward to fit using MCMC.
- ▶ How to pick J ?
- ▶ Can we have more basis functions than observations?

Thank you!